Nishal Stanislaus Silva, *PhD*

ML Research Engineer | Real-Time & Edge Inference | Audio, Sensor & Time-Series

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Professional Summary

PhD researcher whose doctoral work is, at its core, hardware-aware ML: designing and validating models under a fixed latency, compute, and power envelope on embedded ARM hardware. Headline result: a real-time audio pattern detector running at 14 ms inference on a Raspberry Pi 4, F1 0.76, 31.4% CPU utilization – 74x faster than the 1038 ms DTW baseline it replaced (JAES 2026). Built MIRaaS, a UDP edge-to-server offload architecture that characterizes exactly where accuracy, latency, and compute trade off when deciding what runs on-edge versus off-edge (accepted, IEEE IS2 2026). Track record of frozen-protocol benchmark design, ablation studies, and quantified accuracy/latency/CPU trade-off reporting across five years of applied ML on embedded and industrial hardware.

Core Competencies

Hardware-Aware ML & Edge Optimization: ARM CPU and power profiling; real-time inference under sub-30 ms budgets; embedded Linux deployment (Raspberry Pi 4); model compression, quantization, and ONNX/TFLite familiarity; frozen-protocol benchmark and ablation design; latency/accuracy/CPU trade-off quantification; edge-vs-server offload architecture. **Also:** real-time systems engineering, sensor/time-series pattern detection, computer vision at manufacturing scale, ML pipeline ownership (acquisition through deployment and monitoring).

Technical Stack

C++17, Python, TensorFlow, Keras, PyTorch, scikit-learn, ONNX, TFLite (familiarity), embedded Linux, Raspberry Pi/ARM, Docker, CI/CD, AWS, SQL, ETL pipelines, Git, pytest.

Education

PhD, Information Engineering and Computer Science – University of Trento, Italy 2021 - 2025
Thesis: "Embedded Real-time Musical Pattern Detection for Smart Musical Instruments" – hardware-aware model design under fixed latency/compute/power budgets on embedded ARM hardware.
MSc, Telecommunication and Electronic Engineering – Sheffield Hallam University, UK 2016 - 2018
BEng (Hons), Electronic Engineering – Sheffield Hallam University, UK 2013 - 2014

Research and Work Experience

Independent Researcher | *Self-directed, Toronto, Canada* *Jan 2026 - Present*

- Authored two accepted peer-reviewed papers (Smart Drums, JAES; MIRaaS, IEEE IS2 2026) continuing the hardware-aware ML and edge-vs-server characterization line of work.
- Maintain Nebula, a C++17 audio-feature-extraction library with per-feature ARM/Raspberry Pi latency benchmarks (github.com/ns2max/nebula, 13 stars).
- Use AI-assisted development workflows, including Claude Code, to accelerate library and tooling development.

Postdoctoral Researcher | *University of Trento, Trento, Italy* *Jan 2025 - Dec 2025*

- Owned the full ML pipeline for streaming audio and IMU/gesture data on the EU Horizon EIC Pathfinder project MUSMET: acquisition, DSP/feature engineering, training and evaluation, edge deployment on Raspberry Pi 4, and monitoring.
- Designed and shipped a real-time audio pattern detector at 14 ms inference, F1 0.76, 31.4% CPU on Raspberry Pi 4 – 74x faster than the 1038 ms DTW baseline it replaced (JAES 2026).

- Designed frozen-protocol benchmark suites and ablations (RNN vs. DTW; audio vs. MIDI; pattern-set sizes 1/3/10) quantifying accuracy, latency, and CPU trade-offs, with a written record of negative results.
- Built MIRaaS, a UDP edge-to-server offload architecture characterizing latency/compute trade-offs for on-edge vs. off-edge inference decisions (accepted, IEEE IS2 2026).

Visiting Researcher | *McGill University (IDMIL / CIRMMT), Montreal, Canada* *May 2024 - Aug 2024*

- Built a self-powered smart electric guitar (IMU + audio) with gesture fusion via a hierarchical FSM, reaching F1 0.78 on a 500-event corpus (IEEE IS2 2025).
- Profiled compute and power draw to validate real-time feasibility on self-powered embedded hardware – power-budget hardware-aware engineering.

Visiting Researcher | *University of Visual and Performing Arts, Colombo, Sri Lanka* *Jan 2024 - Mar 2024*

- Conducted a human-centred evaluation of musicians' perception of smart instruments with embedded pattern detection (IEEE I3DA 2025).

Technical Consultant | *Forestpin, Colombo, Sri Lanka* *Aug 2020 - Feb 2021*

- Built ML and statistical pipelines for financial forensics, compliance monitoring, and risk detection on large structured enterprise data.

Research Engineer | *MAS Holdings, Colombo, Sri Lanka* *Jan 2015 - Jul 2020*

- Delivered end-to-end computer-vision and IoT systems for manufacturing at scale over 5.5 years, raising operational efficiency by up to 300%.
- Ran real-time IoT ETL pipelines across sites, deploying C++/Python inference on embedded hardware, and introduced CI/CD and code review practices.
- Automated multilingual care-label quality control (OpenCV, conveyor + PLC reject integration), cutting inspection time by 99.5%.

Publications

Silva, N. & Turchet, L. (2026). Real-Time Audio Pattern Detection for Smart Musical Instruments. *J. Audio Eng. Soc.* 74(3). Silva, N. & Turchet, L. (2026, in press). Smart Drums. *J. Audio Eng. Soc.* Silva, N. & Turchet, L. (2026, accepted). MIRaaS: Latency Characterization of an Edge-to-Server Architecture. *IEEE IS2*. Silva, N. & Turchet, L. (2022). A Training-Free SSIM-Based Method for Onset Detection, achieving 95% detection with no training data. *DAFx*.

Highlights

Finalist, MIDI Innovation Awards 2023 (Prototypes & Non-Commercial Software), for a real-time smart-instrument prototype; recognized by the Government Medical Officers' Association of Sri Lanka (2020) for a COVID-19 remote vitals-monitoring deployment; peer reviewer for IEEE Access and the Journal of the National Science Foundation of Sri Lanka; programme committee member for IEEE IS2 and IEEE I3DA; supervised two BSc students.